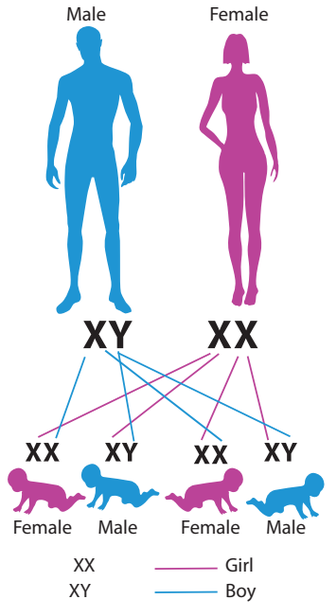
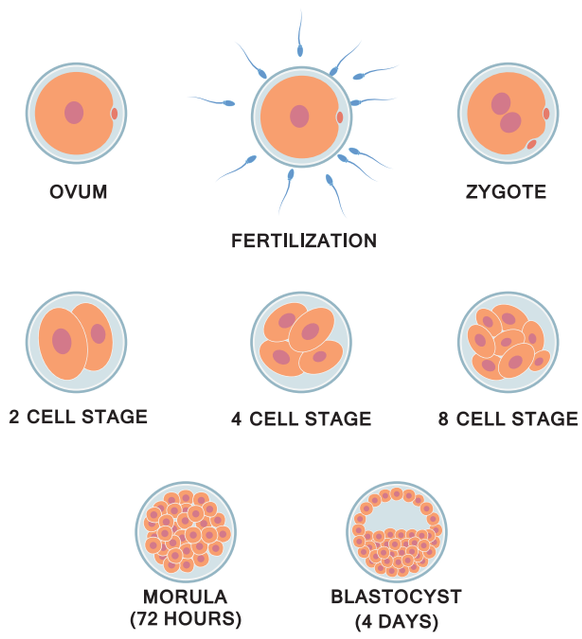
Sex determination



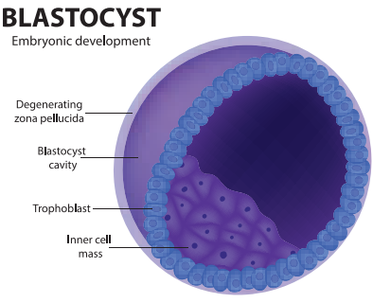


- The cells which have the potency to give rise to any types of cells in the body are called stem cells.
- Embryo cells act as stem cells upto the age of 6 days and have the ability to form any part of the body.

Stages of Clevation



Structure of Blastocyst



- After fertilization cleavage of zygote takes place. It consists of repeated mitotic divisions of the zygot which results in rapid increase in the number of the cells. These smaller embryonic cells are called blastomeres. This normally occurs in the uterine tube.
- The embryo with 8 to 16 blastomeres is called a morula. The Morula reaches the uterine cavity at this stage. Spherical Morula is formed about 3 days after fertilization.
- The morula divides further as it moves further into the uterus and transforms into blastocyst.
- The blastomeres in the blastocyst are arranged into an outer layer called trophoblast and inner mass of cells attached to trophoblast is called as inner cell mass.
- By 7th day, Trophoblast is differentiated into 2 layers: Cytotrophoblast, inner layer, mononucleated mitotically active cells. Syncytiotrophoblast (outer multinucleated mass, with indistinct cell boundary).
- By 8th day the blastocyst is superficially embedded in the compact layer of the endometrium.
- By the 10th day the blastocyst is completely buried in the uterine lining, known as “Implantation” or “embedding" “some

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